

YEAR 7 • UNIT 1 • EXTRA HOMEWORK

The Two That Do the Work

Multiplication and division • the engine room of Unit 1

Name: _____

Class: _____

Date given: _____

Date due: _____

Why this sheet exists

Everything in Unit 1 sits on top of two operations. A **factor** is found by **dividing**, factor **pairs** by **multiplying**, and the **HCF** simplifies a fraction by **dividing** top and bottom. So the faster and more surely you can multiply and divide, the easier the whole unit gets. That is all this sheet is: **extra practice on the two operations that do the work.**

Do it without a calculator, and show your working — working you can see is working you can check.

Part 1 • The facts worth knowing by heart

E1 Fill in the missing number

These are the ones that come up again and again in factors and divisibility. Aim to know them without counting.

(a) $6 \times 7 =$ _____

(f) $8 \times 8 =$ _____

(k) $8 \times 12 =$ _____

(b) $7 \times 8 =$ _____

(g) $12 \times 7 =$ _____

(l) $9 \times 12 =$ _____

(c) $8 \times 9 =$ _____

(h) $11 \times 9 =$ _____

(m) $7 \times 9 =$ _____

(d) $9 \times 6 =$ _____

(i) $9 \times 9 =$ _____

(n) $11 \times 11 =$ _____

(e) $7 \times 7 =$ _____

(j) $12 \times 12 =$ _____

(o) $7 \times 11 =$ _____

E2 Multiply

Set each one out properly and show your working in the box.

(a) Two digits by one digit.

34×6	47×8	58×7	93×4

(b) Three digits by one digit.

236×7	418×6	364×8

(c) Two digits by two digits.

23×14	36×25	48×17

Part 2 · Division

E3 Divide – these all come out exactly

A check that costs five seconds

When you finish a division, **multiply your answer back**. If $72 \div 6 = 12$, then 12×6 must come to 72. If it does not, you have just caught your own mistake.

(a) $72 \div 6 =$ _____

(e) $156 \div 12 =$ _____

(b) $96 \div 4 =$ _____

(f) $245 \div 5 =$ _____

(c) $91 \div 7 =$ _____

(g) $372 \div 4 =$ _____

(d) $126 \div 9 =$ _____

(h) $651 \div 3 =$ _____

E4 Divide – these do not

Word help – the words that describe what is left

exactly nothing is left over. $72 \div 6 = 12$ exactly.

remainder what is left over when it does not come out exactly. We write $53 \div 6 = 8$ r 5.

goes into “6 goes into 24 four times” means $24 \div 6 = 4$.

left over the everyday English for a remainder.

Write each answer as a whole number and a remainder, like this: $53 \div 6 = 8 \text{ r } 5$.

(a) $74 \div 8 =$ _____

(d) $137 \div 5 =$ _____

(b) $100 \div 7 =$ _____

(e) $191 \div 6 =$ _____

(c) $85 \div 9 =$ _____

(f) $250 \div 8 =$ _____

Part 3 • Putting them together

E5 Find the missing number

Multiplying and dividing undo each other

If $7 \times 9 = 63$, then $63 \div 7 = 9$ **and** $63 \div 9 = 7$. Every multiplication fact is also **two** division facts — which is why Part 1 is worth knowing by heart.

(a) $7 \times$ _____ $= 63$

(e) $6 \times$ _____ $= 84$

(b) _____ $\times 8 = 96$

(f) _____ $\times 9 = 117$

(c) $54 \div$ _____ $= 6$

(g) $132 \div$ _____ $= 11$

(d) _____ $\div 7 = 8$

(h) _____ $\div 4 = 25$

E6 Use it: is it a factor?

Straight out of Homework 3

A number is a **factor** of another if it divides into it **exactly**. So the way to find out is to **do the division and look at the remainder**.

Do each division, then circle **yes** or **no**. **Two of these you have already worked out on this sheet** — look back rather than starting again.

Question	The division	Is it a factor?
Is 6 a factor of 84?		yes / no
Is 8 a factor of 92?		yes / no
Is 9 a factor of 126?		yes / no
Is 7 a factor of 100?		yes / no

E7 Word problems**Word help – which operation does the sentence want?**

each / every	usually multiply. 6 boxes with 24 in each is 6×24 .
share equally between	divide.
how many are left	the remainder.
altogether / in total	the whole amount, once you have finished.

Write the calculation first, then the answer.

1. **The markers.** Mr Bowen buys **6 boxes** of markers. There are **24 markers** in each box. How many markers does he have altogether?

2. **The paper.** Mr Bowen shares **96 sheets of paper** equally between **8 students**. How many sheets does each student get?

3. **The pizzas.** Mr Bowen orders **8 identical pizzas** for the class. Each pizza is cut into **12 slices**. There are **29 students**. Every student gets the same whole number of slices, and Mr Bowen eats the rest. How many slices does each student get, and how many does Mr Bowen eat?